

Solucion tarea 9

1. Descargamos Hive

Para ello lo instalamos en el escritorio /home/Kali/Desktop/ e instalamos la versión 4.0.1 de hive compatible con Java 11

```
kali@kali: ~/Desktop
$ wget https://archive.apache.org/dist/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz
--2026-02-23 05:02:29-- https://archive.apache.org/dist/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 208.149.134.204, 65.180.204.189
Connecting to archive.apache.org (archive.apache.org):208.149.134.204... failed: Connection timed out.
Connecting to archive.apache.org (archive.apache.org):65.180.204.189... connected.
HTTP request sent, awaiting response... 200 OK
Length: 408225272 (388M) [application/gzip]
Saving to: 'apache-hive-4.0.1-bin.tar.gz'

apache-hive-4.0.1-bin.tar.gz 100% [=====] 389.31M 520KB/s in 14m 43s

2026-02-23 05:18:27 (451 KB/s) = 'apache-hive-4.0.1-bin.tar.gz' saved [408225272/408225272]

$ tar -xvzf apache-hive-4.0.1-bin.tar.gz
apache-hive-4.0.1-bin/hadoop-practiced-recursos
apache-hive-4.0.1-bin/hadoop-practiced-recursos
```

2. Descomprimos Hive

```
kali@kali: ~/Desktop
$ tar -xvzf apache-hive-4.0.1-bin.tar.gz
apache-hive-4.0.1-bin/conf/hive-log4j2.properties.template
apache-hive-4.0.1-bin/conf/hive-exec-log4j2.properties.template
apache-hive-4.0.1-bin/conf/beeline-log4j2.properties.template
apache-hive-4.0.1-bin/conf/llap-daemon-log4j2.properties.template
apache-hive-4.0.1-bin/conf/llap-cli-log4j2.properties.template
apache-hive-4.0.1-bin/conf/parquet-logging.properties
apache-hive-4.0.1-bin/hcatalog/share/doc/hcatalog/README.txt
apache-hive-4.0.1-bin/NOTICE
apache-hive-4.0.1-bin/RELEASE_NOTES.txt
apache-hive-4.0.1-bin/LICENSE
apache-hive-4.0.1-bin/licenses.xml
apache-hive-4.0.1-bin/licenses/APACHE-1.1.txt
apache-hive-4.0.1-bin/licenses/APACHE-2.0
apache-hive-4.0.1-bin/licenses/BSD-2-CLAUSE
apache-hive-4.0.1-bin/licenses/BSD-3-CLAUSE
apache-hive-4.0.1-bin/licenses/CDDL-1.1.html
apache-hive-4.0.1-bin/licenses/EPL-1.0
apache-hive-4.0.1-bin/licenses/MIT
apache-hive-4.0.1-bin/licenses/bouncy-castle-licence-licence.html
apache-hive-4.0.1-bin/licenses/bsd-2-clause-license.txt
apache-hive-4.0.1-bin/licenses/bsd-3-clause-license.txt
apache-hive-4.0.1-bin/licenses/bsd-licence-licence.html
apache-hive-4.0.1-bin/licenses/bsd-license-licence.txt
apache-hive-4.0.1-bin/licenses/bsd-license.html
apache-hive-4.0.1-bin/licenses/cddl-+-gplv2-with-classpath-exception-license.txt
apache-hive-4.0.1-bin/licenses/eclipse-public-license-1.0-epl-v10.html
apache-hive-4.0.1-bin/licenses/eclipse-public-license-v.-2.0-epl-2.0.txt
apache-hive-4.0.1-bin/licenses/eclipse-public-license-v2.0-epl-v20.html
apache-hive-4.0.1-bin/licenses/edl-1.0-edl-v10.html
apache-hive-4.0.1-bin/licenses/go-license-licence.html
apache-hive-4.0.1-bin/licenses/mozilla-public-license-version-2.0-2.0.html
apache-hive-4.0.1-bin/licenses/new-bsd-license-licence.txt
apache-hive-4.0.1-bin/licenses/the-apache-software-license-version-2.0-license.txt
apache-hive-4.0.1-bin/licenses/the-mit-license-licence.txt
apache-hive-4.0.1-bin/licenses/the-sax-license-copying.html
apache-hive-4.0.1-bin/licenses/the-w3c-license-java-binding.txt
apache-hive-4.0.1-bin/licenses/wtfpl-.html
apache-hive-4.0.1-bin/examples/files/test_hcube_2.db/
apache-hive-4.0.1-bin/examples/files/2000_cols_data.csv
apache-hive-4.0.1-bin/examples/files/3col_data.txt
apache-hive-4.0.1-bin/examples/files/4col_data.txt
apache-hive-4.0.1-bin/examples/files/5col_data.txt
apache-hive-4.0.1-bin/examples/files/AvroPrimitiveInList.parquet
apache-hive-4.0.1-bin/examples/files/AvroSingleFieldGroupInList.parquet
apache-hive-4.0.1-bin/examples/files/HiveGroup.parquet
apache-hive-4.0.1-bin/examples/files/HiveRequiredGroupInList.parquet
apache-hive-4.0.1-bin/examples/files/MultiFieldGroupInList.parquet
```

La carpeta apache-hive-4.0.1-bin indica que se ha descomprimido correctamente

```
(kali㉿nodo1)-[~/Desktop]
$ ls
apache-hive-4.0.1-bin  apache-hive-4.0.1-bin.tar.gz  archivo.txt  hadoop  practica8  recursos
```

3. Renombramos la carpeta

```
Session  Actions  Edit  View  Help

(kali㉿nodo1)-[~/Desktop]
$ mv apache-hive-4.0.1-bin hive

(kali㉿nodo1)-[~/Desktop]
$ ls
apache-hive-4.0.1-bin.tar.gz  archivo.txt  hadoop  hive  practica8  recursos

(kali㉿nodo1)-[~/Desktop]
$
```

4. Configuramos las variables de entorno

Abrimos el archivo .bashrc y escribimos al final del todo las variables de entorno de hive

```
GNU nano 8.6 /home/kali/.bashrc

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -l'
alias la='ls -A'
alias l='ls -CF'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc)
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

# Java
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH

# Hadoop
export HADOOP_HOME=/home/kali/Desktop/hadoop
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

export HADOOP_CLASSPATH=$JAVA_HOME/lib/tools.jar

# HIVE
export HIVE_HOME=/home/kali/Desktop/hive
export PATH=$PATH:$HIVE_HOME/bin
```

5. Creamos el archivo empleados_data.txt

```
kali@nodo1: ~  
Session Actions Edit View Help  
GNU nano 8.6 empleados_data.txt  
Michael|Montreal,Toronto|Male,30|DB:80|Product:Developer,Lead  
Will|Montreal|Male,35|Perl:85|Product:Lead,Test:Lead  
Shelley|New York|Female,27|Python:80|Test:Lead,COE:Architect  
Lucy|Vancouver|Female,57|Sales:89,HR:94|Sales:Lead
```

6. Creamos las tablas internas y vemos su contenido

Para ello primero entramos a Hive

```
kali@nodo1: ~  
Session Actions Edit View Help  
hive  
SLF4J: Class path contains multiple SLF4J bindings.  
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.  
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]  
SLF4J: Class path contains multiple SLF4J bindings.  
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.  
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]  
Beeline version 4.0.1 by Apache Hive  
beeline>
```

Creamos la base de datos y la tabla interna

```
# jdbc:hive2://<host>:<port> --hiveconf <key>=<value>  
0: jdbc:hive2://> CREATE DATABASE IF NOT EXISTS curso_bigdata;  
26/02/23 18:43:16 [HiveServer2-Background-Pool: Thread-57]: WARN exec.DDLTask: metastore.warehouse.external.dir is not set, falling back to metastore.warehouse.dir. This could cause external tables to use to managed tablespace.  
No rows affected (2.972 seconds)  
0: jdbc:hive2://> USE curso_bigdata;  
No rows affected (0.221 seconds)  
0: jdbc:hive2://> CREATE TABLE empleados_interna (  
+-----+> name STRING,  
+-----+> work_place ARRAY<STRING>,  
+-----+> sex_age STRUCT<sex:STRING, age:INT>,  
+-----+> skills_score MAP<STRING, INT>,  
+-----+> depart_title MAP<STRING, ARRAY<STRING>>  
+-----+> )  
+-----+> COMMENT 'Tabla interna con datos complejos'  
+-----+> ROW FORMAT DELIMITED  
+-----+> FIELDS TERMINATED BY ','  
+-----+> COLLECTION ITEMS TERMINATED BY ';' ;  
No rows affected (1.113 seconds)  
0: jdbc:hive2://>
```

```
No rows affected (0.033 seconds)  
0: jdbc:hive2://> SHOW DATABASES;  
+-----+  
| database_name |  
+-----+  
| curso_bigdata |  
| default       |  
+-----+  
2 rows selected (0.455 seconds)  
0: jdbc:hive2://> USE curso_bigdata;  
No rows affected (0.084 seconds)  
0: jdbc:hive2://> SHOW TABLES;  
+-----+  
| tab_name      |  
+-----+  
| empleados_interna |  
+-----+  
1 row selected (0.326 seconds)  
0: jdbc:hive2://>
```

Ver el contenido de la tabla

```
0: jdbc:hive2://> SELECT * FROM empleados_internal;
26/02/24 06:36:19 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN calcite.RelOptHiveTable: No Stats for curso_bigdata@empleados_internal, Columns: work_place, depart_title, name, skills_score, sex_age
No Stats for curso_bigdata@empleados_internal, Columns: work_place, depart_title, name, skills_score, sex_age
26/02/24 06:36:19 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN optimizer.SimpleFetchOptimizer: Table curso_bigdata@empleados_internal is external table, falling back to filesystem scan.
+-----+-----+-----+-----+-----+
| empleados_internal.name | empleados_internal.work_place | empleados_internal.sex_age | empleados_internal.skills_score | empleados_internal.depart_title |
+-----+-----+-----+-----+-----+
| Michael                 | ["Montreal", "Toronto"]       | [{"sex": "Male", "age": 30}] | [{"DB": 80}]                     | [{"Product": ["Developer"], "Lead": null}] |
| Will                   | ["Montreal"]                  | [{"sex": "Male", "age": 35}] | [{"Perl": 85}]                   | [{"Product": ["Lead"], "Test": ["Lead"]} ] |
| Shelley                 | ["New York"]                  | [{"sex": "Female", "age": 27}] | [{"Python": 88}]                 | [{"Test": ["Lead"], "COC": ["Architect"]} ] |
| Lucy                   | ["Vancouver"]                | [{"sex": "Female", "age": 57}] | [{"Sales": 89, "HR": 94}]        | [{"Sales": ["Lead"]} ] |
+-----+-----+-----+-----+-----+
```

Acceder a un array (primera ciudad de trabajo)

```
0: jdbc:hive2://> SELECT name, work_place[0] as ciudad_principal FROM empleados_internal;
26/02/24 06:37:07 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN calcite.RelOptHiveTable: No Stats for curso_bigdata@empleados_internal, Columns: work_place, name
No Stats for curso_bigdata@empleados_internal, Columns: work_place, name
26/02/24 06:37:07 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN optimizer.SimpleFetchOptimizer: Table curso_bigdata@empleados_internal is external table, falling back to filesystem scan.
+-----+-----+
| name | ciudad_principal |
+-----+-----+
| Michael | Montreal |
| Will | Montreal |
| Shelley | New York |
| Lucy | Vancouver |
+-----+-----+
```

Acceder a una struct (empleados mayores de 30)

```
0: jdbc:hive2://> SELECT name, sex_age.age FROM empleados_internal WHERE sex_age.age > 30;
26/02/24 06:38:06 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN calcite.RelOptHiveTable: No Stats for curso_bigdata@empleados_internal, Columns: name, sex_age
No Stats for curso_bigdata@empleados_internal, Columns: name, sex_age
26/02/24 06:38:06 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN optimizer.SimpleFetchOptimizer: Table curso_bigdata@empleados_internal is external table, falling back to filesystem scan.
+-----+-----+
| name | age |
+-----+-----+
| Will | 35 |
| Lucy | 57 |
+-----+-----+
2 rows selected (0.635 seconds)
0: jdbc:hive2://>
```

Acceder a un map (puntuación en habilidad DB)

```
0: jdbc:hive2://> SELECT name, skills_score['DB'] as nota_db FROM empleados_internal;
26/02/24 06:38:42 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN calcite.RelOptHiveTable: No Stats for curso_bigdata@empleados_internal, Columns: name, skills_score
No Stats for curso_bigdata@empleados_internal, Columns: name, skills_score
26/02/24 06:38:42 [3fb05483-4b7b-4a04-b6aa-d8d091c24dc2 main]: WARN optimizer.SimpleFetchOptimizer: Table curso_bigdata@empleados_internal is external table, falling back to filesystem scan.
+-----+-----+
| name | nota_db |
+-----+-----+
| Michael | 80 |
| Will | NULL |
| Shelley | NULL |
| Lucy | NULL |
+-----+-----+
```

En una terminal nueva creamos una carpeta simulando un departamento, subimos el fichero y comprobamos que los cambios han sido correctos

```
hali@nodeh:~$
Session Actions Edit View Help

\\[e]0:[u]0h: \\[u]0:\\[033[;32m\\]---\\[\\[033[;34m\\][u]0\\[\\[033[;32m\\]]-\\[\\[033[0;1m\\][w\\[\\[033[;32m\\]]u\\[\\[033[;32m\\]]-\\[\\[033[;34m\\][B\\[\\[033[0m\\] /home/kali/Desktop/hadoop/bin/hdfs dfs -mkdir -p /empresa/rhhd/datos
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

\\[e]0:[u]0h: \\[u]0:\\[033[;32m\\]---\\[\\[033[;34m\\][u]0\\[\\[033[;32m\\]]-\\[\\[033[0;1m\\][w\\[\\[033[;32m\\]]u\\[\\[033[;32m\\]]-\\[\\[033[;34m\\][B\\[\\[033[0m\\] /home/kali/Desktop/hadoop/bin/hdfs dfs -put /home/kali/empleados_data.txt /empresa/rhhd/datos
JMS/
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

\\[e]0:[u]0h: \\[u]0:\\[033[;32m\\]---\\[\\[033[;34m\\][u]0\\[\\[033[;32m\\]]-\\[\\[033[0;1m\\][w\\[\\[033[;32m\\]]u\\[\\[033[;32m\\]]-\\[\\[033[;34m\\][B\\[\\[033[0m\\] /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /empresa/rhhd/datos/
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Found 1 items
-rw-r--r-- 2 kali supergroup 227 2026-02-24 06:41 /empresa/rhhd/datos/empleados_data.txt

\\[e]0:[u]0h: \\[u]0:\\[033[;32m\\]---\\[\\[033[;34m\\][u]0\\[\\[033[;32m\\]]-\\[\\[033[0;1m\\][w\\[\\[033[;32m\\]]u\\[\\[033[;32m\\]]-\\[\\[033[;34m\\][B\\[\\[033[0m\\]
```

Creamos una tabla externa

```
0: jdbc:hive2://> USE curso_bigdata;
No rows affected (0.046 seconds)
0: jdbc:hive2://>
0: jdbc:hive2://> CREATE EXTERNAL TABLE empleados_external (
  . . . . . > name STRING,
  . . . . . > work_place ARRAY<STRING>,
  . . . . . > sex_age STRUCT<sex:STRING, age:INT>,
  . . . . . > skills_score MAP<STRING, INT>,
  . . . . . > depart_title MAP<STRING, ARRAY<STRING>>
  . . . . . > )
  . . . . . > ROW FORMAT DELIMITED
  . . . . . > FIELDS TERMINATED BY '|'
  . . . . . > COLLECTION ITEMS TERMINATED BY ','
  . . . . . > MAP KEYS TERMINATED BY ':'
  . . . . . > LOCATION '/empresa/rrhh/datos';
No rows affected (0.408 seconds)
0: jdbc:hive2://> █
```

Mostramos los datos de la tabla

```
0: jdbc:hive2://> select * FROM empleados_external;
26/02/24 06:44:40 [3fbd5483-4b7b-4a04-b6aa-d8d891c24dc2 main]: WARN calcite.RelOptHiveTable: No Stats for curso_bigdata@empleados_external, Columns: work_place, depart_title, name, skills_score, sex_age
No Stats for curso_bigdata@empleados_external, Columns: work_place, depart_title, name, skills_score, sex_age
26/02/24 06:44:40 [3fbd5483-4b7b-4a04-b6aa-d8d891c24dc2 main]: WARN optimizer.SimpleFetchOptimizer: Table curso_bigdata@empleados_external is external table, falling back to filesystem scan.

+-----+-----+-----+-----+-----+
| empleados_external.name | empleados_external.work_place | empleados_external.sex_age | empleados_external.skills_score | empleados_external.depart_title |
+-----+-----+-----+-----+-----+
| Michael                 | ["Montreal","toronto"]         | {"sex":"Male","age":30}     | {"DB":100}                       | [{"Product":["Developer"],"Lead":null}] |
| Will                   | ["Montreal"]                  | {"sex":"Male","age":35}     | {"Perl":35}                     | [{"Product":["Lead"],"Test":["Lead"]}    |
| Shelley                 | ["New York"]                  | {"sex":"Female","age":27}   | {"Python":80}                   | [{"Test":["Lead"],"COE":["Architect"]}   |
| Lucy                   | ["Vancouver"]                 | {"sex":"Female","age":57}   | {"Sales":89,"HR":94}            | [{"Sales":["Lead"]}                     |
+-----+-----+-----+-----+-----+
4 rows selected (0.343 seconds)
0: jdbc:hive2://> █
```

Borramos la tabla interna

```
0: jdbc:hive2://> DROP TABLE empleados_internal;
No rows affected (0.852 seconds)
0: jdbc:hive2://> █
```

Comprobamos si los datos se han borrado que es lo que queremos

```
\\[v]0[0]u0[h]: \\w[a]\\[\\033[32m\\]---\\[\\033[1;34m\\]u0[h]\\[\\033[32m\\]--\\[\\033[0;1m\\]w\\[\\033[32m\\]w\\[\\033[32m\\]---\\[\\033[1;34m\\]$(\\033[0m\\) /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /user/hive/warehouse/curso_bigdata.db/
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

\\[v]0[0]u0[h]: \\w[a]\\[\\033[32m\\]---\\[\\033[1;34m\\]u0[h]\\[\\033[32m\\]--\\[\\033[0;1m\\]w\\[\\033[32m\\]w\\[\\033[32m\\]---\\[\\033[1;34m\\]$(\\033[0m\\) █
```

Borramos la tabla externa

```
0: jdbc:hive2://> DROP TABLE empleados_external;
No rows affected (0.13 seconds)
0: jdbc:hive2://> █
```

Comprobamos si los datos NO se han borrado que es lo que queremos

```
\\[v]0[0]u0[h]: \\w[a]\\[\\033[32m\\]---\\[\\033[1;34m\\]u0[h]\\[\\033[32m\\]--\\[\\033[0;1m\\]w\\[\\033[32m\\]w\\[\\033[32m\\]---\\[\\033[1;34m\\]$(\\033[0m\\) /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /empresa/rrhh/datos/
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Found 1 items
-rw-r--r-- 2 kali supergroup 227 2026-02-24 06:41 /empresa/rrhh/datos/empleados_data.txt

\\[v]0[0]u0[h]: \\w[a]\\[\\033[32m\\]---\\[\\033[1;34m\\]u0[h]\\[\\033[32m\\]--\\[\\033[0;1m\\]w\\[\\033[32m\\]w\\[\\033[32m\\]---\\[\\033[1;34m\\]$(\\033[0m\\) █
```